

GeoWeave: Learning Reliable Cross-View Dependencies for Feed-Forward 3D Reconstruction

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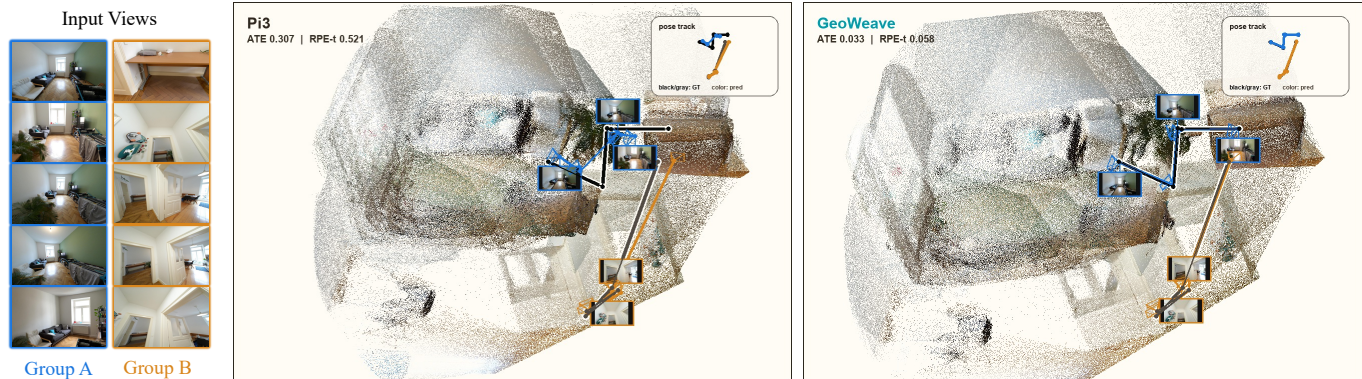


Fig. 1. Reliable cross-view dependencies for weak-overlap reconstruction. Given two local view groups with limited overlap, Pi3 produces less consistent camera poses and 3D structure. GeoWeave learns to select reliable cross-view dependencies, yielding a more coherent feed-forward 3D reconstruction.

Recent feed-forward multi-view reconstruction models replace explicit correspondence search with learned transformer-based fusion for joint camera and 3D structure inference. This flexibility makes geometric reasoning rely heavily on data-driven token-level interactions. Yet multi-view reconstruction is not a generic token-mixing problem: not every cross-view relation provides useful evidence, and geometry-relevant signals may be obscured by feature-similar relations with limited geometric support. We introduce GeoWeave, which addresses this mismatch by scoring candidate cross-view dependencies and confining attention to the most reliable relations. Each global fusion block uses a lightweight learned scorer to rank candidate relations for each token, then applies the original attention aggregation within the selected set. Training follows a two-stage procedure: first, dense-attention distillation teaches the scorer a stable ranking from the original model’s attention maps; second, the full reconstruction model is trained

end to end with scorer-guided attention under geometric supervision. On standard pose and point-reconstruction benchmarks, GeoWeave matches the corresponding baselines overall and brings clear gains on complex real-world datasets. We further design two challenging evaluation settings to test whether dependency selection improves reconstruction under low overlap or distracting context; compared with Pi3, which can collapse in these settings, GeoWeave maintains substantially stronger robustness.

CCS Concepts: • **Computing methodologies** → **Reconstruction**; *Computer vision*; *Computer graphics*.

Additional Key Words and Phrases: 3D reconstruction, attention, VGGT

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1 Introduction

Recent feed-forward multi-view reconstruction models estimate camera geometry and dense 3D structure by aggregating information across views with transformer architectures [Keetha et al. 2025; Leroy et al. 2024; Wang et al. 2025a, 2024; Yang et al. 2025]. Rather than constructing explicit matches, triangulating points, and refining them with bundle adjustment [Hartley and Zisserman 2004; Schönberger and Frahm 2016; Schönberger et al. 2016; Triggs et al. 2000], they learn multi-view representations from which geometry can be predicted in a single forward pass. This data-driven design

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is flexible across scenes and view configurations, with cross-view geometric evidence emerging from learned token-level interactions rather than explicit correspondence constraints [Sun et al. 2025; Wang and Xu 2025].

In these models, token-level interactions are typically realized through dense global self-attention. This low-bias design gives each query broad access to tokens across input views without imposing a predefined matching structure. Yet multi-view reconstruction is not a generic token-mixing problem: not every cross-view relation provides useful evidence, since many candidates may arise from weakly related or ambiguous cross-view relations. When attention is operated over the full candidate set, geometry-relevant signals may be obscured by feature-similar relations that provide limited geometric support. This suggests a form of cross-view attention that preserves broad access while operating on reconstruction-relevant relations.

To this end, we present GeoWeave, a feed-forward multi-view reconstruction model with scorer-guided cross-view attention. GeoWeave factors cross-view fusion into learned dependency selection and standard attention-based aggregation. A lightweight scorer first ranks and retains candidate relations for each query token, producing a filtered cross-view context. Within this context, GeoWeave computes standard full-dimensional attention using the original queries, keys, and values. This design preserves the flexible, data-driven nature of transformer fusion, while encouraging each update to draw from a more selective set of reconstruction-relevant cross-view dependencies.

Training the entire GeoWeave model from scratch is unstable, because unreliable early scorer rankings can narrow the cross-view evidence available to attention and drive the model toward suboptimal dependency patterns. We therefore train GeoWeave in two stages. In the first stage, we distill relation preferences from the pretrained dense global attention into the scorer by aligning the scorer distribution with dense attention maps, giving the scorer a stable initialization over the full candidate pool. In the second stage, we enable scorer-guided attention and optimize the full GeoWeave model end to end with the geometric reconstruction objective. Importantly, we retain the distribution alignment loss in this stage, but apply it within the selected context; this provides an auxiliary learning signal for dependency selection without prescribing explicit geometric matches.

We evaluate GeoWeave along two axes: general reconstruction performance and robustness under challenging view configurations. On standard pose and point-reconstruction benchmarks, GeoWeave remains competitive with the corresponding baselines despite restricting each query to a selected dependency set, and it brings clear gains on challenging real-world evaluations such as ETH3D [Schöps et al. 2017] and RealEstate10K [Zhou et al. 2018]. We further design two controlled settings to test whether dependency selection improves reconstruction robustness under low view overlap and distracting inputs; results on Waymo [Sun et al. 2020] and ScanNet++ [Yeshwanth et al. 2023] show that GeoWeave maintains stable reconstructions where dense-attention baselines are more easily disrupted, as illustrated by the weak-overlap example in Fig. 1.

In summary, our contributions are:

- We propose GeoWeave, a scorer-guided cross-view attention mechanism for feed-forward multi-view reconstruction that factors fusion into learned dependency selection and standard full-dimensional attention-based aggregation.
- We introduce a two-stage scorer-training strategy with distribution alignment losses: dense-attention distillation initializes the scorer over the full candidate pool, and we then train GeoWeave end to end with a geometric reconstruction objective, using selected-context alignment as an auxiliary supervision signal.
- We provide standard benchmark results and targeted robustness protocols showing that GeoWeave improves challenging real-world reconstruction cases and stabilizes cross-view fusion under weak overlap or distracting context.

2 Related Work

2.1 Feed-Forward Multi-View Reconstruction

Classical 3D reconstruction pipelines estimate camera geometry and scene structure through explicit geometric stages, including multi-view geometry [Hartley and Zisserman 2004], structure-from-motion [Schönberger and Frahm 2016], bundle adjustment [Triggs et al. 2000], and dense multi-view stereo [Schönberger et al. 2016]. Learning-based multi-view stereo methods such as MVSNet [Yao et al. 2018], CasMVSNet [Gu et al. 2020], TransMVSNet [Ding et al. 2022], and MVSFormer++ [Cao et al. 2024] introduced end-to-end depth inference from posed images by constructing cost volumes, aggregating multi-view features, and regularizing depth hypotheses with CNN or transformer architectures.

Recent feed-forward reconstruction models move toward a more unified formulation in which geometry is predicted directly from image observations. DUST3R [Wang et al. 2024] regresses dense point maps from image pairs without requiring known calibration, MAST3R [Leroy et al. 2024] strengthens this formulation for image matching, and subsequent models such as Fast3R [Yang et al. 2025], VGGT [Wang et al. 2025a], and MapAnything [Keetha et al. 2025] extend feed-forward reconstruction to larger multi-view sets and joint prediction of camera parameters, depth, and 3D structure. GeoWeave follows this feed-forward reconstruction paradigm and focuses on how cross-view dependencies are selected inside transformer-based fusion.

2.2 Global Attention in Feed-Forward Reconstruction

Global self-attention has become a central mechanism for multi-view fusion in recent feed-forward reconstruction models such as Fast3R [Yang et al. 2025], VGGT [Wang et al. 2025a], and MapAnything [Keetha et al. 2025], where tokens from all input views interact in a shared transformer representation. This broad access supports joint camera and structure inference, but also makes global fusion a central design point. Recent work mainly revises this pathway for scalability: VGGT-Long [Deng et al. 2025] and Stream3R [Lan et al. 2025] introduce chunked or streaming processing for long sequences, while Faster VGGT [Wang et al. 2025b], FlashVGGT [Wang and Xu 2025], and AVGGT [Sun et al. 2025] reduce global-attention cost through sparsity, descriptor compression, or attention conversion.

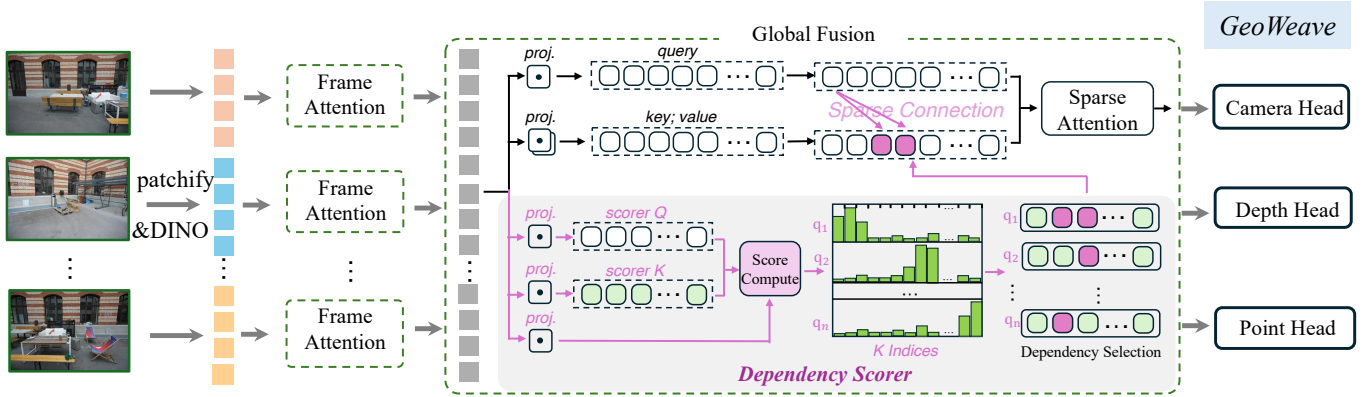


Fig. 2. GeoWeave architecture. After per-view tokenization and frame attention, tokens are fused by global transformer blocks. GeoWeave augments each global fusion block with a dependency scorer, which computes query-conditioned scores over candidate keys and selects a dependency set for each query. The selected key indices define the context for attention aggregation, while the attention values are still computed with the original full-dimensional queries, keys, and values.

GeoWeave is complementary to these efforts but targets a different aspect of global fusion. Existing approaches mainly reorganize the global-attention computation through sequence chunking, streaming schedules, token compression, or attention approximation. In contrast, GeoWeave focuses on the dependency structure underlying global fusion: which cross-view relations should provide evidence for reconstruction, rather than assuming a fixed attention domain over all candidate relations.

3 GeoWeave

GeoWeave is built on the alternating-attention transformer used in VGGT-style feed-forward reconstruction [Wang et al. 2025a]. Given image tokens produced by a visual encoder, the backbone alternates between frame-wise self-attention within each image and global self-attention over tokens from all input views. As shown in Fig. 2, GeoWeave leaves the tokenization, frame-wise attention, and prediction heads unchanged, and modifies only the global fusion operation. In each global block, a dependency scorer assigns query-conditioned scores to candidate keys, selects a context for each query, and standard full-dimensional attention aggregates values within the selected context.

3.1 Preliminaries: Alternating-Attention Backbone

Following VGGT [Wang et al. 2025a], we use an alternating-attention backbone for feed-forward multi-view reconstruction. Given V input images, a visual encoder converts each image into T patch tokens. We denote by $\mathbf{x}_{v,t} \in \mathbb{R}^d$ the embedding of the t -th token from view v . The backbone alternates frame-wise self-attention, which processes tokens within each image, with global self-attention, which jointly processes tokens from all input views.

For a global self-attention layer, tokens from all views are concatenated into a single sequence of length $L = VT$:

$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_L] \in \mathbb{R}^{L \times d}. \quad (1)$$

The layer computes

$$\begin{aligned} \mathbf{Q} &= \mathbf{X} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{X} \mathbf{W}_V, \\ \text{Attn}(\mathbf{X}) &= \text{Softmax}\left(\frac{\mathbf{Q} \mathbf{K}^T}{\sqrt{d}}\right) \mathbf{V}. \end{aligned} \quad (2)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V$ are learned projections and the softmax is applied row-wise over all L keys. This dense global attention gives each token broad access to all tokens from all input views. GeoWeave keeps this backbone structure, but changes the global attention layer by selecting the candidate relations used for each query before applying full-dimensional attention aggregation.

3.2 Scorer-Guided Global Fusion

Standard global attention forms each query’s aggregation over the entire multi-view token sequence. Because not every cross-view relation provides reconstruction-relevant support for a given query, full-pool aggregation can make geometry-relevant signals compete with weak or ambiguous candidates. GeoWeave preserves the full-dimensional QKV attention computation, but introduces a dependency scorer that ranks candidate keys before aggregation. The selected keys define a query-specific context, within which attention is computed using the original queries, keys, and values.

Dependency scorer. In each global fusion block, the dependency scorer estimates a query-key score map using a lightweight scoring branch. Its role is not to replace the attention logits used for feature aggregation, but to provide a stable ordering of candidate keys before attention is applied. We therefore project tokens to lower-dimensional scorer queries and keys $\tilde{\mathbf{q}}_i, \tilde{\mathbf{k}}_j \in \mathbb{R}^{d_s}$, where $d_s \ll d$, while keeping the full-dimensional QKV branch unchanged for aggregation. The scorer also predicts a source-side gate g_j , a scalar that modulates the contribution of candidate key j to the ranking. For readability, we write the gated positive similarity as

$$s_{ij} = g_j \text{ReLU}\left(\frac{\tilde{\mathbf{q}}_i^T \tilde{\mathbf{k}}_j}{\sqrt{d_s}}\right). \quad (3)$$

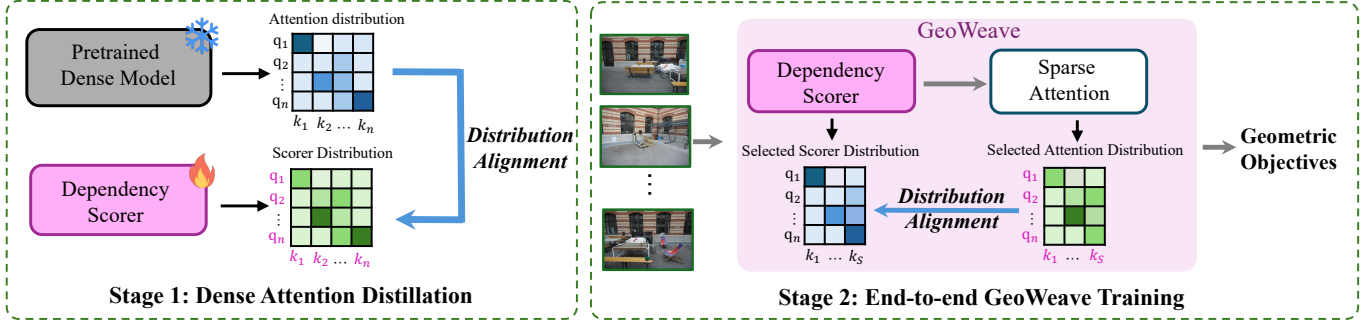


Fig. 3. Two-stage training of GeoWeave. In Stage 1, a pretrained dense model is frozen and its dense attention distribution provides a distillation target for the dependency scorer. This dense-attention alignment initializes the scorer over the full candidate pool. In Stage 2, the full GeoWeave model is trained end to end with geometric objectives, while a selected-context alignment loss keeps the scorer distribution consistent with the attention distribution within the selected dependency set.

The ReLU suppresses negative affinities, while the source-side gate adjusts each candidate key’s contribution to the score.

Dependency selection. The dependency scores are used to form a query-conditioned context for global attention. For each query token $\mathbf{x}_i$, GeoWeave retains a dependency set $\mathcal{N}(i)$ containing k candidate keys. Let $\mathbf{q}_i$, $\mathbf{k}_j$, and $\mathbf{v}_j$ denote the original full-dimensional attention projections. Attention within the selected dependency set is computed as

$$\begin{aligned} \mathcal{N}(i) &= \text{TopK}_j(s_{ij}, k), \\ \alpha_{ij} &= \frac{\exp(\mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d})}{\sum_{m \in \mathcal{N}(i)} \exp(\mathbf{q}_i^\top \mathbf{k}_m / \sqrt{d})}, \quad j \in \mathcal{N}(i), \\ y_i &= \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{v}_j. \end{aligned} \quad (4)$$

The scores s_{ij} are used only to determine which keys enter $\mathcal{N}(i)$; once the context is selected, the attention weights α_{ij} are computed from the original full-dimensional queries and keys. The selected set therefore represents data-driven support relations for feature aggregation, rather than explicit geometric correspondences. In practice, we compute $\mathcal{N}(i)$ once per token and reuse the same selected indices across attention heads, keeping the dependency set query-conditioned while leaving multi-head aggregation unchanged.

Drop-in replacement. GeoWeave leaves the rest of the base reconstruction model unchanged: frame attention, token resolution, positional encoding, and task heads are preserved. We replace only the dense global attention operation with scorer-guided selected-context attention. This keeps the method compatible with the original camera, depth, and point prediction objectives.

3.3 Training

Training the dependency scorer from scratch is unstable because early inaccurate rankings can restrict the evidence exposed to attention, making it harder for the model to learn better selections. We therefore train GeoWeave in two stages (Fig. 3).

Stage 1: dense-attention distillation. We first freeze the pretrained dense model and use its global attention distributions as supervision

for the dependency scorer. The scorer is trained to reproduce the relation preferences implied by dense attention over the full candidate pool. Concretely, for each global attention block and each query token i , the frozen model yields dense attention weights $\mathbf{a}_i^{\text{dense}} \in \mathbb{R}^L$ over all keys.¹ Given scorer logits $\mathbf{s}_i = [s_{i1}, \dots, s_{iL}]$ and temperature τ , we minimize a dense score alignment loss:

$$\mathcal{L}_{\text{align}}^{\text{dense}} = \sum_{i=1}^L \text{KL}(\text{sg}[\mathbf{a}_i^{\text{dense}}] \parallel \text{Softmax}(\mathbf{s}_i / \tau)), \quad (5)$$

where $\text{sg}[\cdot]$ denotes stop-gradient. This initializes the scorer with a stable ranking over the full candidate pool before hard selection is activated.

Stage 2: End-to-End GeoWeave training. After Stage 1, the scorer provides a stable initial ranking over the full candidate pool. Stage 2 activates selected-context attention and optimizes the complete GeoWeave model with the original geometric reconstruction objective $\mathcal{L}_{\text{geo}}$, which supervises camera parameters, depth maps, and 3D point outputs. Under this computation, each query aggregates features only from its selected dependency set $\mathcal{N}(i)$, making dependency selection part of the reconstruction path.

Although $\mathcal{L}_{\text{geo}}$ provides task-level supervision through the final reconstruction outputs, it gives only indirect feedback to the scorer that decides which candidate relations enter $\mathcal{N}(i)$. We therefore retain distribution alignment as dependency-level supervision for the scorer. Concretely, within the selected context, the attention weights computed from the original query and key projections serve as a local target for the scorer. Aligning to this target encourages the scorer to assign higher scores to relations used by the aggregation branch, so the selected dependency sets are refined together with the reconstruction model during end-to-end optimization. For each query, we align the scorer distribution with these attention weights over $\mathcal{N}(i)$:

$$\mathbf{p}_i^{\text{sel}} = \text{Softmax}(\{\ell_{ij}\}_{j \in \mathcal{N}(i)}), \quad \mathbf{q}_i^{\text{sel}} = \text{Softmax}(\{s_{ij}/\tau\}_{j \in \mathcal{N}(i)}), \quad (6)$$

$$\mathcal{L}_{\text{align}}^{\text{sel}} = \sum_{i=1}^L \text{KL}(\mathbf{p}_i^{\text{sel}} \parallel \mathbf{q}_i^{\text{sel}}), \quad (7)$$

¹In multi-head attention, we use the head-averaged attention distribution as the target.

Table 1. Camera pose estimation on standard benchmarks. We report AUC@30 in percentage; higher is better. Bold numbers indicate the better result within each backbone pair.

Method	RE10K	CO3Dv2	ETH3D	DTU64
VGGT	84.9	89.9	85.5	85.3
GeoWeave-VGGT	86.9	89.5	85.1	86.1
Pi3	88.0	88.2	84.4	95.0
GeoWeave-Pi3	87.7	85.5	87.6	95.5

Table 2. Point reconstruction results. We report accuracy (Acc), completeness (Comp), and normal consistency (NC). Lower is better for Acc and Comp, while higher is better for NC. Bold numbers indicate the better result within each backbone pair.

Method	DTU			ETH3D			7Scenes		
	Acc	Comp	NC	Acc	Comp	NC	Acc	Comp	NC
VGGT	1.185	2.227	0.694	0.275	0.323	0.849	0.0228	0.0383	0.7249
GeoWeave-VGGT	1.165	2.240	0.687	0.245	0.271	0.860	0.0186	0.0331	0.6722
Pi3	1.152	1.795	0.668	0.187	0.210	0.872	0.0210	0.0319	0.6741
GeoWeave-Pi3	1.200	1.917	0.665	0.166	0.188	0.876	0.0167	0.0285	0.6734

The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{geo}} + \lambda \mathcal{L}_{\text{align}}^{\text{sel}}, \quad (8)$$

with a scalar weight λ . This selected-context alignment keeps dependency scoring calibrated to the attention branch used for aggregation, while $\mathcal{L}_{\text{geo}}$ optimizes the geometric predictions.

4 Experiments

4.1 Experimental Setup

4.1.1 Training. We evaluate GeoWeave on two feed-forward reconstruction backbones, VGGT and Pi3, resulting in GeoWeave-VGGT and GeoWeave-Pi3. Stage 1 distillation uses the corresponding official dense model as teacher, i.e., VGGT [Wang et al. 2025a] for GeoWeave-VGGT and Pi3 [Wang et al. 2026] for GeoWeave-Pi3. Across Stage 1 and Stage 2, each variant follows the official preprocessing, view sampling, and supervision format of its corresponding backbone; Stage 2 also uses the corresponding original geometric reconstruction objective.

Our training data use the full 17-dataset mixture: BlendedMVS train [Yao et al. 2020], Hypersim train and val [Roberts et al. 2021], MVS-Synth/GTAV [Huang et al. 2018], Replica [Straub et al. 2019], TartanAir [Wang et al. 2020], Virtual KITTI 2 [Caban et al. 2020], Aria Synthetic Environments (ASE) [Avetisyan et al. 2024], Aria Digital Twin (ADT) [Pan et al. 2023], MegaDepth [Li and Snavely 2018], ScanNet++ [Yeshwanth et al. 2023], Taskonomy [Zamir et al. 2018], WildRGB-D train [Xia et al. 2024], Map-Free train [Arnold et al. 2022], CO3Dv2 train [Reizenstein et al. 2021], Mapillary Street-Level Sequences [Warburg et al. 2020], and DL3DV [Ling et al. 2024]. We sample datasets in proportion to scene counts and process each variant with the official data pipeline of its source backbone.

4.1.2 Evaluation Protocol. Following VGGT [Wang et al. 2025a], our evaluation centers on camera pose estimation and point reconstruction, where the quality of cross-view fusion is directly reflected in the consistency of the recovered geometry. The selected benchmarks cover large-scale video, object-centric scenes, and calibrated multi-view reconstruction settings.

For camera pose, we use RealEstate10K [Zhou et al. 2018] and CO3Dv2 [Reizenstein et al. 2021]. We also evaluate ETH3D [Schöps et al. 2017] and DTU64 [Jensen et al. 2014]. RealEstate10K and CO3Dv2 evaluate camera recovery under large-scale video and object-centric settings, while ETH3D and DTU64 emphasize structured multi-view geometry. We report AUC@30 as the primary pose metric, using the same evaluator and input views for all methods.

For point reconstruction, we use the same DTU and ETH3D datasets, together with 7-Scenes [Shotton et al. 2013]. DTU and ETH3D follow the standard VGGT point-cloud protocol, while 7-Scenes provides an indoor reconstruction setting with different scene statistics. We report accuracy, completeness, and normal consistency.

Our primary baselines are the original VGGT [Wang et al. 2025a] and Pi3 [Wang et al. 2026] models. For fair comparison, all baseline results are reproduced using the official model checkpoints under the dataset settings of the corresponding original papers. For each benchmark, all compared models use the same input protocol, preprocessing, task definition, and evaluation script.

4.1.3 Implementation Details. Guided by the layer-wise analysis in AVGGT [Sun et al. 2025], which shows that cross-view alignment behavior is concentrated in the middle global attention layers, we apply GeoWeave to the corresponding global fusion blocks rather than to all blocks. GeoWeave-VGGT applies the dependency scorer to global layers 9–19, while GeoWeave-Pi3 applies it to global layers 9–17. We retain TopK=1024 keys for each query and set the scorer dimension to $d_s = 64$.

Both stages are trained on 32 RTX 5090 GPUs with mixed precision in fp16. Stage 1 runs for 40k steps with a cosine-decayed learning rate initialized at 10^{-3} . Stage 2 trains for 80k steps using a peak learning rate of 10^{-5} with linear warm-up and cosine decay to 10^{-8} . We use AdamW with weight decay 5×10^{-2} . The selected-context alignment weight is set to $\lambda = 0.2$ for GeoWeave-VGGT and $\lambda = 10$ for GeoWeave-Pi3. We analyze the computational cost of the scorer/indexer in Supplementary Sec. 5, including parameter count, inference and training time, and GPU memory usage.

4.2 Standard Benchmark Results

Table 1 reports camera pose accuracy on the selected benchmarks. GeoWeave restricts each query to a Top-K dependency set, substantially narrowing the scope of cross-view token interaction from the full candidate pool to a selected subset of relations. Despite this restriction, it remains competitive with the corresponding dense-attention backbones. This behavior supports the view that feed-forward transformer reconstruction can be interpreted in part as a learned search over geometry-relevant cross-view dependencies, rather than as uniform aggregation over all token relations.

The advantages are most visible on challenging real-world geometry benchmarks. GeoWeave-VGGT improves camera estimation

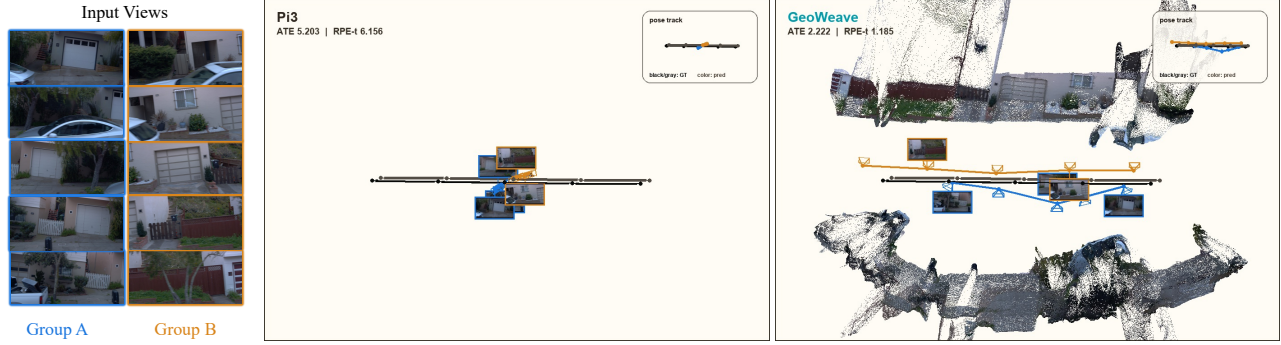


Fig. 4. Weak-overlap Waymo visualization. In this side-camera setting, Pi3 produces a collapsed camera trajectory, whereas GeoWeave-Pi3 maintains a coherent trajectory prediction under the same input. Together with the ScanNet++ teaser in Fig. 1, this visualization illustrates how scorer-guided dependency selection improves camera consistency when cross-view overlap is limited.

on RE10K and DTU64, while GeoWeave-Pi3 improves pose accuracy on ETH3D and DTU64. Table 2 shows a similar pattern for point reconstruction: both variants consistently improve accuracy, completeness, and normal consistency on ETH3D, and both improve accuracy and completeness on 7-Scenes. The ETH3D Top-K visualization in Fig. 5 is consistent with this interpretation: for a query token, the selected regions concentrate on related structures across overlapping views rather than covering the full token pool uniformly. These results indicate that scorer-guided dependency selection can preserve the flexibility of feed-forward reconstruction while making cross-view fusion more focused on geometry-relevant evidence. Unless otherwise specified, all Top-K visualizations use the dependency scorer from layer 15 of GeoWeave-Pi3.

4.3 Evaluating Cross-View Dependency Selection

To evaluate whether dependency selection improves robustness under challenging view configurations, we study cases in which valid cross-view evidence is sparse or interleaved with visually plausible distractors. We primarily evaluate camera pose because it directly reflects cross-view consistency. Complementary point, normal, and depth metrics are reported in Supplementary Sec. 3. We use ScanNet++ [Yeshwanth et al. 2023] and Waymo [Sun et al. 2020], whose real captured scenes provide complex indoor and outdoor

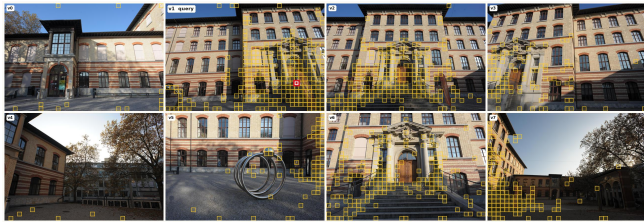


Fig. 5. Top-K dependency selection on ETH3D. The red dot marks the query token, and yellow boxes indicate the regions selected for attention. The selected regions concentrate on visually and geometrically related structures across overlapping views, providing a qualitative view of how GeoWeave organizes cross-view evidence under standard benchmark inputs.

geometry for controlled analysis. The **weak-overlap view group** experiment (Sec. 4.3.1) constructs two local view groups with limited overlap and evaluates whether selected dependencies preserve both within-group support and sparse cross-group bridge relations. The **distractor context** experiment (Sec. 4.3.2) keeps the evaluated reconstruction target fixed while adding visually similar but geometrically inconsistent context images, measuring how much this context perturbs the prediction. These evaluations are not intended as new benchmarks; they provide controlled evidence about how scorer-guided selection affects cross-view fusion.

4.3.1 Weak-Overlap View Groups. To test dependency selection under limited cross-view overlap, we design a weak-overlap setting with two local view groups. Views have substantial overlap within each group, while cross-group overlap is limited and may provide little direct geometric evidence. In such cases, the relative



Fig. 6. Top-K dependency selection for weak-overlap view groups. The red dot marks the query token, and yellow boxes indicate regions selected for attention. The two examples use queries from different view groups. Selected regions provide broad support within the query's local group while preserving sparse cross-group connections around overlapping scene regions, matching the bridge-dependency structure of the weak-overlap setting.



Fig. 7. Distractor context construction on Waymo. The evaluated target is a clean six-frame prefix, while the input also includes four cross-scene context images that are visually similar but geometrically inconsistent. Evaluation is performed only on the clean prefix, so the measured degradation reflects the influence of distractor context on the same reconstruction target.

Table 3. Weak-overlap 10-view evaluation. Each input contains two 5-view groups; all frames are evaluated after a single Sim(3) alignment. Δ ATE is Pi3 minus GeoWeave-Pi3, so positive values indicate improvement.

Dataset	n	Pi3 ATE ↓	GeoWeave-Pi3 ATE ↓	Δ ATE ↑
ScanNet++	12	0.112	0.035	+0.077
Waymo	90	3.921	3.424	+0.497

configuration is underconstrained by the images alone, and reconstruction may partly rely on learned priors. Each input contains 10 views, organized as two 5-view groups. In ScanNet++ [Yeshwanth et al. 2023], the groups are local indoor view clusters with small but nonzero cross-group overlap. In Waymo [Sun et al. 2020], they are synchronized windows from two side cameras. We report pose errors after a single Sim(3) alignment for each tuple, with full tuple construction and evaluation details provided in Supplementary Sec. 1.

Table 3 reports the quantitative results. On ScanNet++, GeoWeave-Pi3 reduces mean ATE from 0.112 to 0.035; the corresponding reconstruction in Fig. 1 shows more coherent camera poses and 3D structure under limited overlap. On Waymo, GeoWeave-Pi3 also lowers mean ATE from 3.921 to 3.424, and Fig. 4 shows that it maintains a coherent camera trajectory where Pi3 collapses. Fig. 6 provides a token-level view of this behavior: for queries from different view groups, selected regions cover strong support inside the query’s local group while also retaining sparse cross-group links around overlapping structures. Together, these results suggest that scorer-guided selection preserves local support while keeping the bridge relations needed for global pose consistency.

4.3.2 Distractor Context. To evaluate robustness to feature-similar context with limited geometric support, we design a Waymo stress test. Each input contains six clean frames that define the evaluated prefix, together with four visually similar but geometrically inconsistent context images from another scene (Fig. 7). The models receive all ten images jointly, while metrics are computed only on the fixed clean prefix; the degradation therefore measures how much the added context perturbs the same reconstruction target.

Table 4. Waymo distractor context stress test. Clean reports performance with clean context, Distractor uses visually similar cross-scene context, and Δ is Distractor minus Clean. The main comparison is degradation, where lower is better.

Method	ATE ↓			RPE-t ↓		
	Clean	Distractor	Δ	Clean	Distractor	Δ
Pi3	0.074	0.140	0.066	0.136	0.218	0.082
GeoWeave-Pi3	0.073	0.111	0.038	0.121	0.165	0.044

Detailed context search, filtering, and evaluation procedures are provided in Supplementary Secs. 2.2 and 2.3.

Table 4 reports both clean-context and distractor-context errors. The clean-context errors are comparable, while GeoWeave-Pi3 has lower error after distractor context is introduced; ATE degradation decreases from 0.066 to 0.038, and translation-RPE degradation decreases from 0.082 to 0.044. Fig. 8 provides a token-level view of this setting: for a query in the clean prefix, selected regions are not dominated by the distractor context. This suggests that scorer-guided selection is less driven by appearance-level plausibility alone, and instead favors cross-view relations that remain useful for geometric reconstruction.

We further assess robustness across multiple cross-group overlap levels and distractor-view counts; detailed results are provided in Supplementary Secs. 1.4 and 2.3. Additional qualitative reconstruction and dependency-selection examples are shown in Figs. 9 and 10.

4.4 Ablation Study

4.4.1 Effect of Dependency Set Size. The Top-K value determines how many scored relations are retained for each query. We ablate this dependency-set size on GeoWeave-Pi3 using ETH3D, keeping all other settings fixed. As shown in Table 5, TopK=1024 gives the best overall balance across pose and point reconstruction. TopK=512 remains competitive for pose but hurts point accuracy and completeness, while TopK=1536 does not further improve the results. We therefore use TopK=1024 as the default setting.



Fig. 8. Top-K dependency selection under distractor context. The red dot marks a query token in the clean prefix, and yellow boxes indicate regions selected for attention. Although the input includes visually similar cross-scene context images, the selected regions remain concentrated around compatible evidence for the query, illustrating how dependency selection limits the influence of distractor context.

Table 5. Top-K ablation on GeoWeave-Pi3 using ETH3D. Pose reports AUC@30 in percentage; point reconstruction reports accuracy (Acc), completeness (Comp), and normal consistency (NC).

Top-K	Pose AUC@30 ↑	Acc ↓	Comp ↓	NC ↑
512	87.1	0.231	0.263	0.867
1024	87.6	0.166	0.188	0.876
1536	83.4	0.181	0.202	0.873

4.4.2 Effect of Stage-2 Alignment. Stage 2 uses selected-context alignment to keep the dependency scorer calibrated to the attention branch used for geometric aggregation. We ablate this component with a variant that starts from the same dense-attention-initialized scorer but removes the Stage-2 alignment loss and keeps the scorer fixed during end-to-end training. Table 6 reports representative results on ETH3D point reconstruction and the Waymo weak-overlap protocol. The ablated model remains stronger than dense Pi3, showing that the selected dependency structure itself is useful. The full GeoWeave-Pi3 further improves ETH3D accuracy and completeness and lowers weak-overlap ATE, indicating that Stage-2 alignment refines the scorer beyond its initialization.

Table 6. Stage-2 selected-context alignment ablation. The ablated model removes the selected-context alignment loss and keeps the scorer fixed during Stage 2. Weak ATE is measured on the Waymo weak-overlap protocol.

Method	ETH3D point			Waymo
	Acc ↓	Comp ↓	NC ↑	Weak ATE ↓
Pi3	0.187	0.210	0.872	3.921
w/o Stage-2 align.	0.180	0.206	0.873	3.509
GeoWeave-Pi3	0.166	0.188	0.876	3.424

4.4.3 Alternative Top-K and Token-Pruning Baselines. Because GeoWeave combines learned scoring with Top-K aggregation, we test whether its robustness can be explained by sparsification alone. We compare against direct and fine-tuned Top-K pruning of Pi3’s dense attention, and include FastVGGT as a generic token-pruning baseline. Table 7 reports results under the same ScanNet++ and Waymo weak-overlap protocols. Direct and fine-tuned Top-K pruning do not recover GeoWeave’s weak-overlap accuracy, while FastVGGT is

Table 7. Comparison with alternative Top-K and token-pruning baselines under the same weak-overlap protocols. Lower is better.

Method	ScanNet++ ATE ↓	Waymo ATE ↓
Pi3-TopK direct	0.185	4.019
Pi3-TopK fine-tuned	0.177	3.997
FastVGGT	0.259	4.726
GeoWeave-Pi3	0.035	3.424

also worse under both protocols. These results demonstrate the advantage of GeoWeave’s learned dependency selection over generic sparsification and token pruning.

5 Conclusion

We presented GeoWeave, a scorer-guided cross-view attention mechanism for feed-forward multi-view reconstruction. GeoWeave learns query-conditioned dependency sets and applies attention aggregation within the selected context, reducing reliance on indiscriminate all-to-all cross-view fusion. The two-stage training procedure first transfers dependency rankings from dense attention and then refines the scorer under geometric supervision. On standard pose and point-reconstruction benchmarks, GeoWeave remains competitive with the corresponding baselines and improves challenging real-world geometry cases. Controlled weak-overlap and distractor-context evaluations further show that learned dependency selection improves reconstruction stability.

Limitations. GeoWeave uses a fixed dependency budget and dense-attention initialization, so its best selection configuration may vary across backbones and scene distributions; the current sparse implementation also incurs additional training and inference costs.

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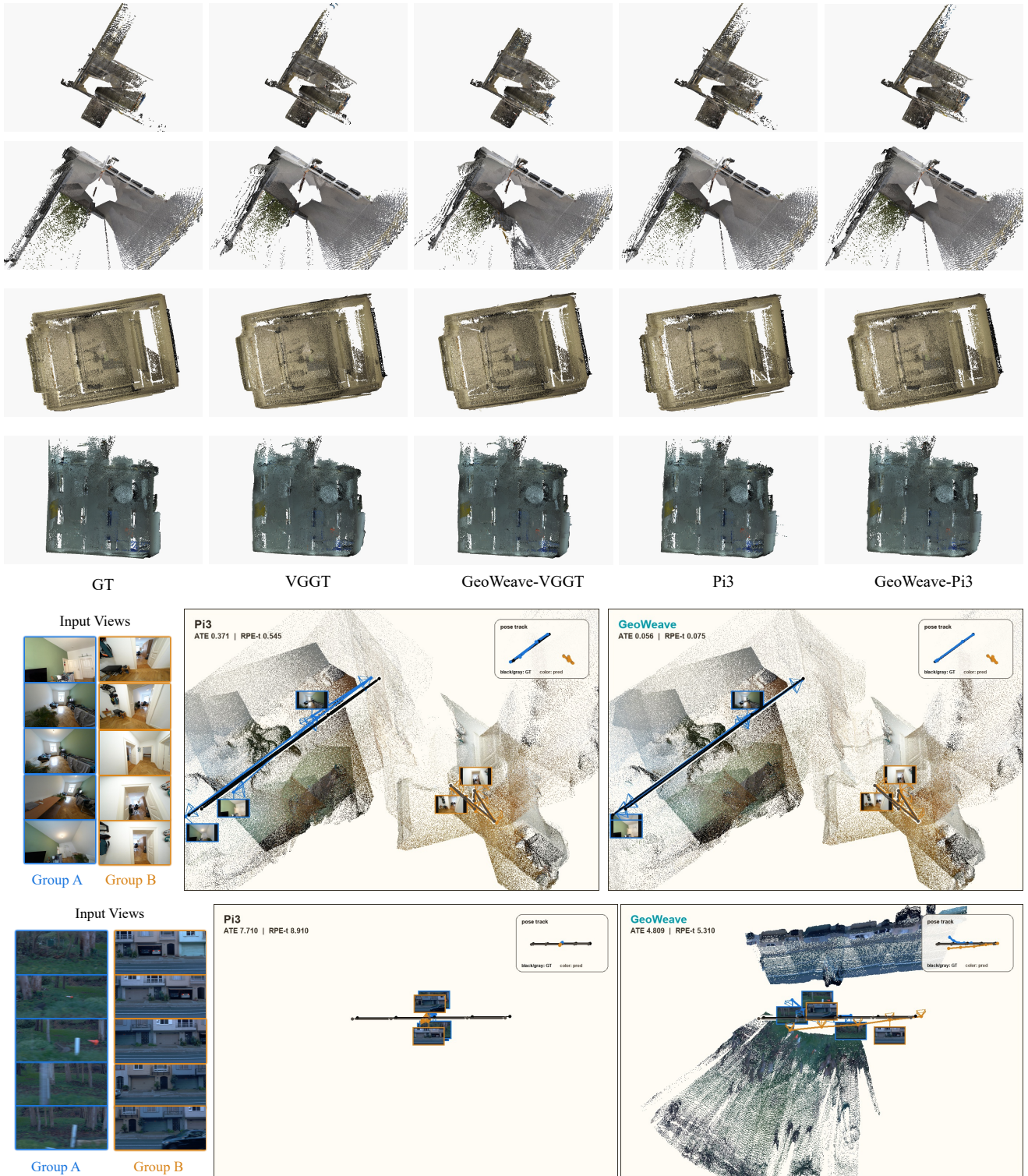


Fig. 9. Additional qualitative results. Top: ETH3D point reconstruction examples, with columns showing ground truth, VGGT, GeoWeave-VGGT, Pi3, and GeoWeave-Pi3. Middle and bottom: additional weak-overlap view-group examples on real captured scenes. The blue and orange input groups have limited cross-group overlap; GeoWeave-Pi3 maintains more coherent camera and structure estimates than Pi3 under the same inputs.



Fig. 10. Additional dependency-selection visualizations. Top: weak-overlap Top-K selection, where selected regions provide support within the query’s local view group while retaining cross-group links around overlapping scene regions. Bottom: additional distractor-context construction and Top-K visualization, where six clean frames define the evaluated prefix and four visually similar cross-scene views provide distractor context.

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